



Rossmoyne Senior High School

Semester One Examination, 2020

Question/Answer booklet

**MATHEMATICS  
METHODS**

**UNIT 1**

**Section Two:**

**Calculator-assumed**

If required by your examination administrator, please  
place your student identification label in this box

WA student number: In figures

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In words

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Circle your Teacher's Name:    **ALVARO**        **BESTALL**        **FRASER-JONES**    **GOH**  
                                                 **KOULIANOS**    **LUZUK**        **RUDLAND**        **TANDAY**

**Time allowed for this section**

Reading time before commencing work: ten minutes  
Working time: one hundred minutes

Number of additional  
answer booklets used  
(if applicable):

|  |
|--|
|  |
|--|

**Materials required/recommended for this section**

***To be provided by the supervisor***

This Question/Answer booklet  
Formula sheet (retained from Section One)

***To be provided by the candidate***

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,  
correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up  
to three calculators approved for use in this examination

**Important note to candidates**

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

See next page

## Structure of this paper

|                                   | Number of questions available | Number of questions to be attempted | Working time (minutes) | Marks available | Percentage of exam |
|-----------------------------------|-------------------------------|-------------------------------------|------------------------|-----------------|--------------------|
| Section One<br>Calculator—free    | 9                             | 9                                   | 50                     | 54              | 36                 |
| Section Two<br>Calculator—assumed | 15                            | 15                                  | 100                    | 96              | 64                 |
|                                   |                               |                                     |                        | 150             | 100                |

## Instructions to candidates

1. The rules for the conduct of Western Australian external examinations are detailed in the *Year 12 Information Handbook 2020*. Sitting this examination implies that you agree to abide by these rules.
2. Answer the questions according to the following instructions.

**Show all your working clearly.** Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat an answer to any question, ensure that you cancel the answer you do not wish to have marked.

It is recommended that you **do not use pencil**, except in diagrams.

3. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
4. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
  - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
  - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question that you are continuing to answer at the top of the page.
5. The Formula Sheet is **not** handed in with your Question/Answer Booklet.

## Section Two: Calculator–assumed

**64% (96 marks)**

This section has **fifteen (15)** questions. Attempt **all** questions. Write your answers in the spaces provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Working time: 100 minutes

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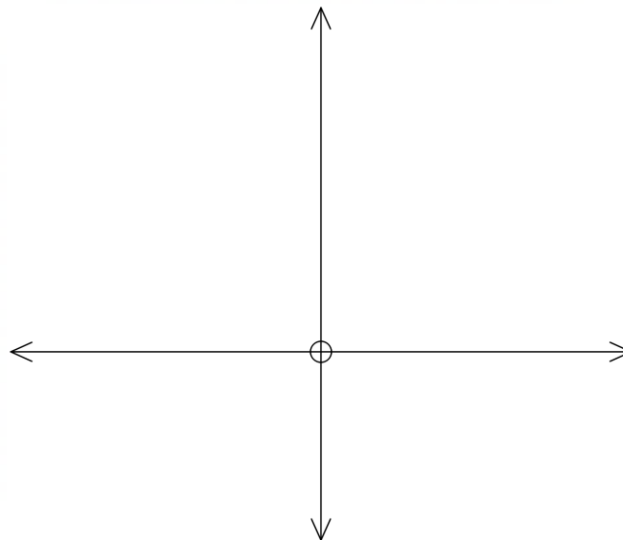
### Question 10 (6 marks)

- (a) A diameter of a circle has endpoints  $P(-4,0)$  and  $Q(16,0)$ . Determine the  $y$  – intercepts of the circle. (3 marks)

- (b) The circumference of a circle ( $C$ ) is directly proportional to the radius ( $r$ ).

- (i) State the exact constant of proportionality. (1 mark)

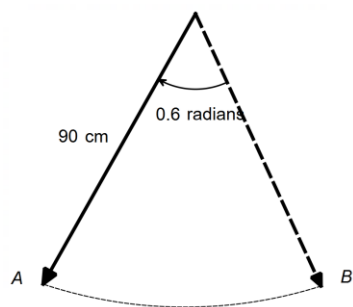
- (ii) Draw a graph on the axes below to show the relationship between  $C$  and  $r$ . Label the axes and all critical features. (2 marks)



See next page

**Question 11** (6 marks)

A pendulum is 90 cm long and swings through an angle of 0.6 radians. The extreme positions of the pendulum are indicated by the points A and B in the diagram.



(a) Find the length of the arc AB. (2 marks)

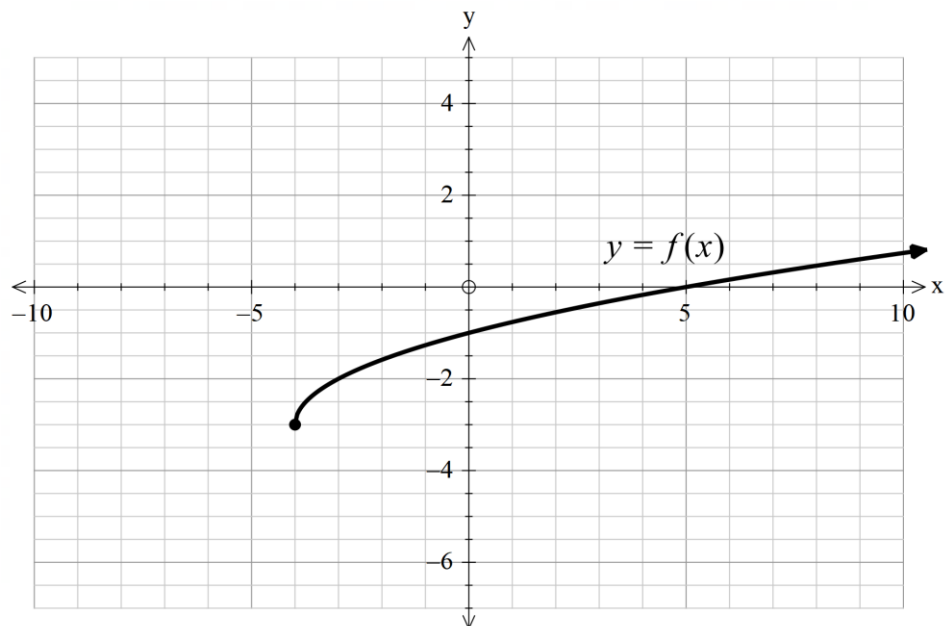
(b) Find the straight line distance between the extreme positions of the pendulum (i.e. the distance from A to B.). Round to 3 decimal places. (2 marks)

(c) Find the area of the sector swept out by the pendulum. (2 marks)

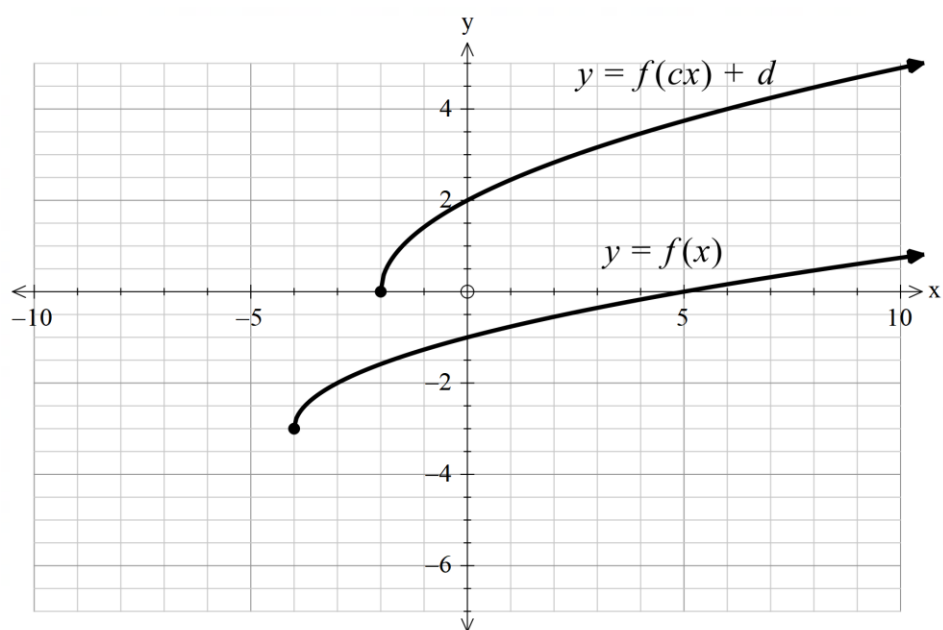
**Question 12 (4 marks)**

The graph of  $y = f(x)$  is drawn below, where  $f(x) = \sqrt{x - a} + b$ .

- (a) Determine the value of the constant  $a$  and the value of the constant  $b$ . (2 marks)



- (b) The graph of  $y = f(cx) + d$  is now graphed on the same axes

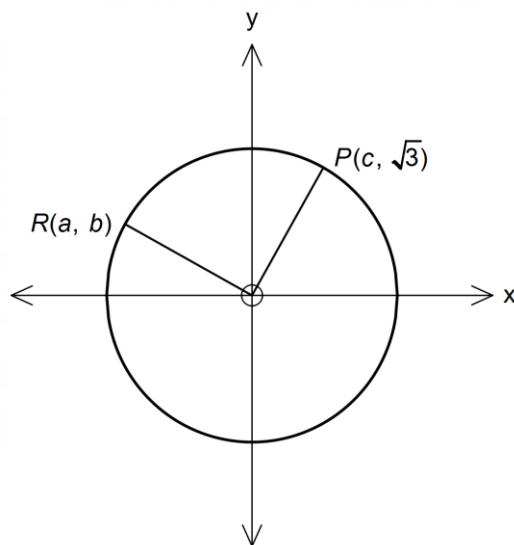


Determine the value of the constant  $c$  and the value of the constant  $d$ . (2 marks)

**Question 13 (7 marks)**

The radius of a circle, with centre  $O$ , is 2 units.  $R(a, b)$  and  $P(c, \sqrt{3})$  are points on the circle.

Let the angle between the positive  $x$  – axis and the line  $OP$  be  $\theta$  and the angle  $ROP$  is  $90^\circ$ .



(a) Show that  $c = 1$  (2 marks)

(b) Determine the value(s) of:

(i)  $\sin \theta$  and hence  $\theta$  . (2 marks)

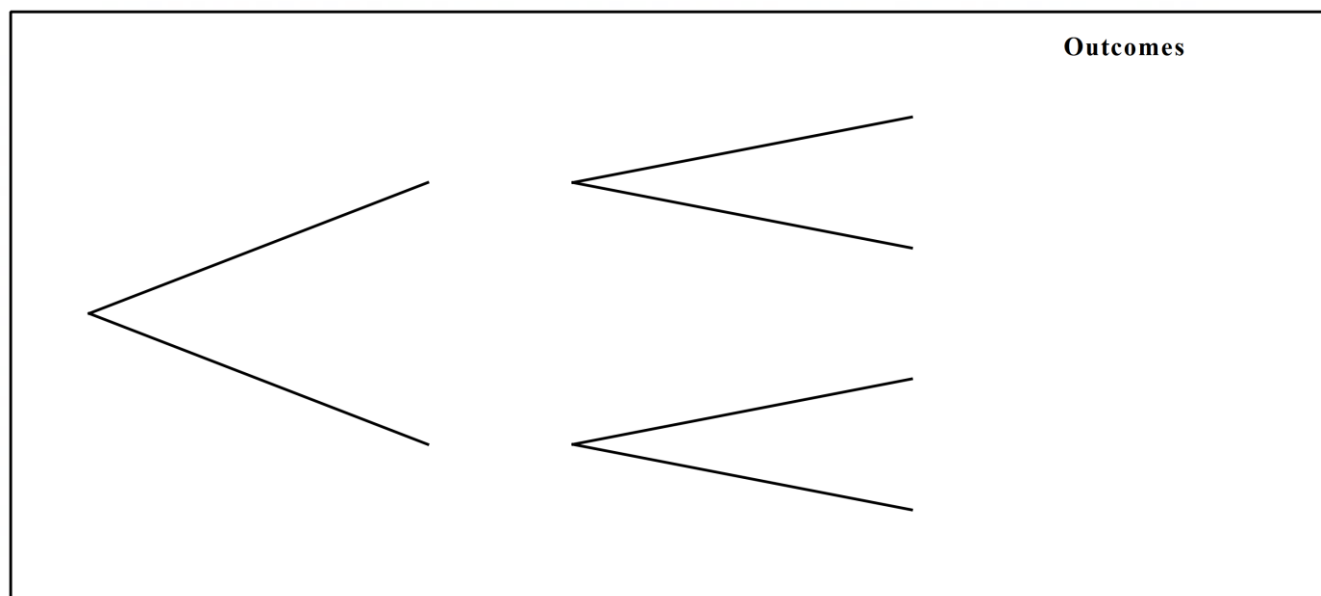
(ii)  $\tan(360^\circ - \theta)$  . (1 mark)

(iii)  $a$  and  $b$ . (2 marks)

**Question 14 (6 marks)**

If it rains on a given day, the probability that it rains the next day is  $\frac{1}{3}$ . If it does not rain on a given day, the probability that it rains the next day is  $\frac{1}{6}$ . The probability that it will rain tomorrow is  $\frac{1}{5}$ .

- (a) Draw a tree diagram to show all the probabilities and outcomes. (3 marks)



- (b) Determine the probability that:

(i) it will rain tomorrow and the next day. (1 mark)

(ii) it will rain the day after tomorrow. (1 mark)

(iii) there will be no rain the next day, given that it rains tomorrow. (1 mark)

**Question 15 (9 marks)**

- (a) Line  $q$  is perpendicular to the line with the equation  $y = 2x - 3$ . Line  $q$  has the same  $x$  - intercept as the line with the equation  $x + 2y = 6$ . Determine the equation of line  $q$ .  
(3 marks)

- (b) For what value(s) of  $k$  is the line through the points  $(3, 2k + 1)$  and  $(8, 4k - 5)$  parallel to the  $x$  - axis.  
(2 marks)

- (c) Consider the quadratic equation  $x^2 - (p + 7)x + p + 87 = 0$  where  $p$  is a negative real number. Solve for  $x$  given that the equation has one distinct real solution.  
(4 marks)



**Question 16 (4 marks)**

It is known that for events A and B,  $P(\bar{A}) = 0.6$ ,  $P(B) = 0.7$  and  $P(A|B) = 0.5$

- (a) Use this information to complete the two-way table. (3 marks)

|              | $P(A)$ | $P(\bar{A})$ | Total |
|--------------|--------|--------------|-------|
| $P(B)$       |        |              |       |
| $P(\bar{B})$ |        |              |       |
| Total        |        |              | 1     |

- (b) Hence or otherwise, determine  $P(\bar{A} \cap \bar{B})$  (1 mark)

**Question 17 (4 marks)**

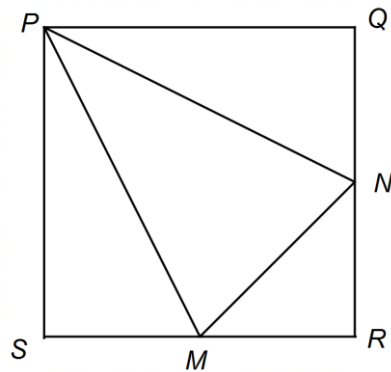
The probabilities that Alf and Bess will attend the next work social are  $P(A) = 0.7$  and  $P(B) = 0.55$  respectively. It is also known that  $P(A \cap B) = 0.45$ .

- (a) Determine  $P(A \cup B)$ . (1 mark)

- (b) Describe, in the context of this question, the event  $(A \cap \bar{B}) \cup (\bar{A} \cap B)$  and calculate the probability that it happens. (3 marks)

**Question 18** (5 marks)

In the diagram, square PQRS has side length 2 cm. Points M and N are midpoints of SR and RQ respectively.



(a) Show that the angle MPN = 0.6435 radians. (3 marks)

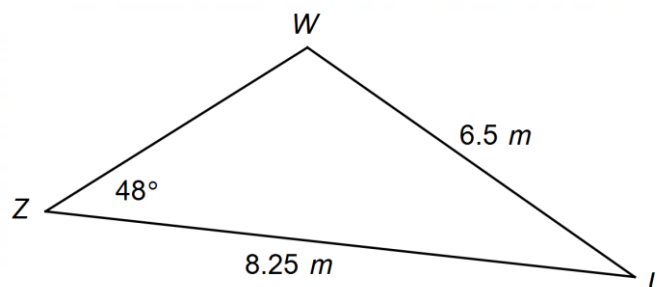
(b) Hence, or otherwise determine the area of  $\Delta$  MPN. (2 marks)

**Question 19 (7 marks)**

- (a) Three aircraft, marked, Z, U and W, are flying in a triangular formation at an air show, at the same speed and altitude. Aircraft Z is 8.25 m from aircraft U which is 6.5 m from aircraft W. The angle WZU formed is  $48^\circ$ .

Determine what distance aircraft W is from aircraft Z?

(3 marks)



- (b) Determine the area of triangle  $PQR$  when  $\angle PQR = 32^\circ$ ,  $\angle PRQ = 114^\circ$  and  $PQ = 37$  cm.

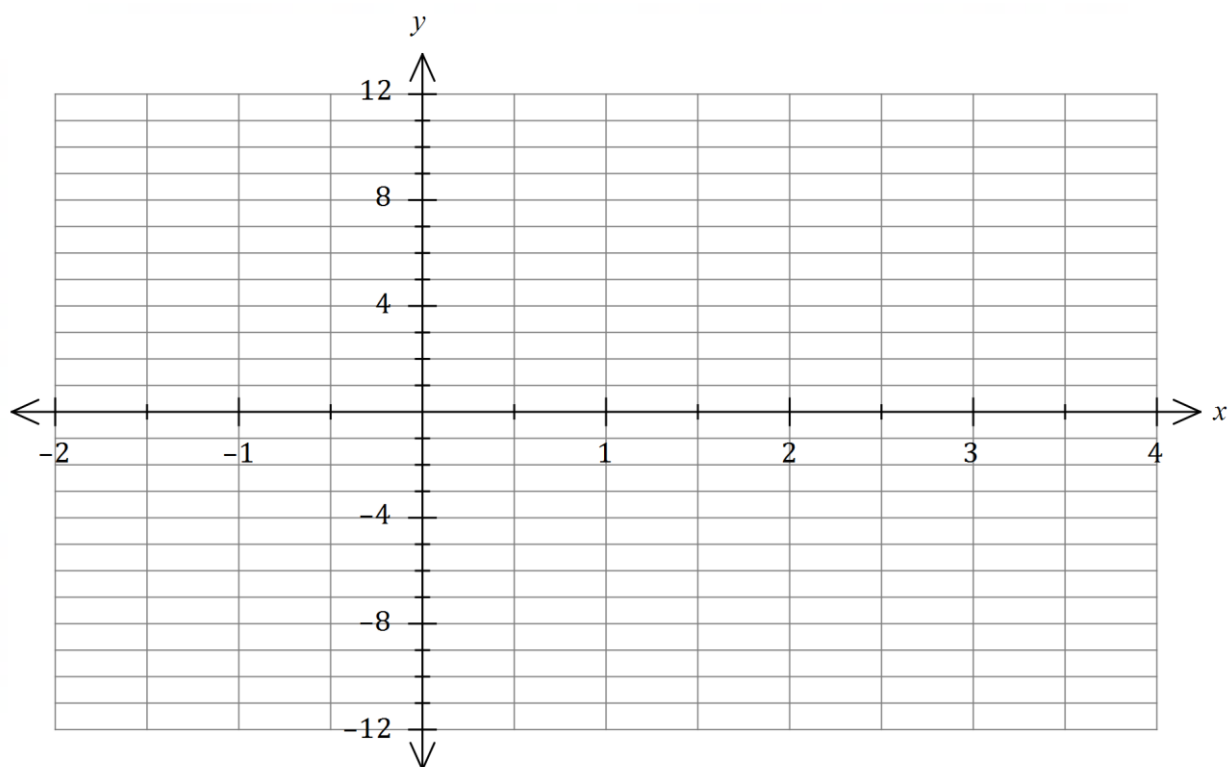
(4 marks)

**Question 20 (8 marks)**

A polynomial of degree 3 passes through the points with coordinates  $(0, 9)$ ,  $(-1, 0)$ ,  $(3, 0)$  and  $(1.5, 0)$ .

- (a) Determine the equation of the polynomial. State in expanded form (3 marks)

- (b) Draw the graph of the polynomial on the axes below, indicating and labelling key features (4 marks)



- (c) Determine the coordinates of the point where concavity changes. (1 mark)  
Give your answer to an accuracy of 4 decimal places.

**Question 21 (6 marks)**

Two events are such that  $P(X) = 0.2$ ,  $P(Y) = 0.5$  and  $P(Y|X) = 0.1$ .

Determine the probability that

(a) both events occur. (2 marks)

(b) at least one event occurs. (2 marks)

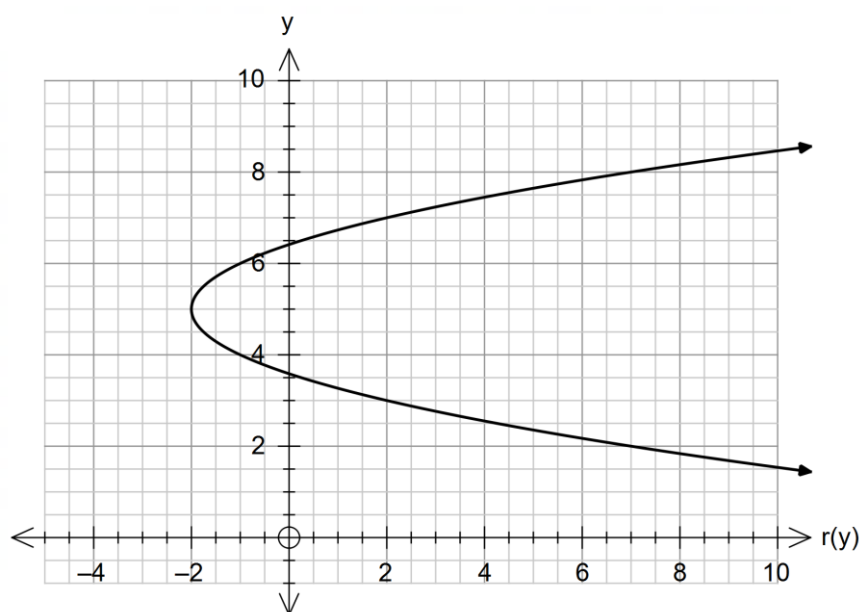
(c) neither event occurs. (1 mark)

(d)  $X$  occurs given that  $Y$  has occurred. (1 mark)

**Question 22 (6 marks)**

(a) (i) State the equation of the graph below:

(2 marks)



(ii) Apply the vertical line test to the graph above and explain its significance. (2 marks)

(b) A straight line makes an angle of  $60^\circ$  with the positive  $x$ -axis and passes through the point with coordinates  $(0, -6)$ . Determine the exact equation of the line. (2 marks)

**Question 23 (10 marks)**

The manager of a private bus company has worked out a model for the relationship between the number of passengers carried each week and the profit, (in tens of \$), the company makes.

If  $n$  is the number of passengers carried, the profit is given by  $P(n) = 2n - 2000 - \left(\frac{n}{100}\right)^2$ .

- (a) Can the relation  $P(n)$  be described as a function. Justify your answer. (1 mark)
- (b) Determine the minimum number of passengers needed per week for the company to make a profit. (1 mark)
- (c) Determine the maximum profit made according to this model and how many passengers are carried during the week to achieve this profit. (2 marks)

- (d) (i) Determine the simplified expression for  $P(n + 1) - P(n)$ , and explain what this expression represents. (4 marks)

- (ii) How many passengers would the bus company carry in a week if the profit made by one extra passenger was 50 cents? (2 marks)

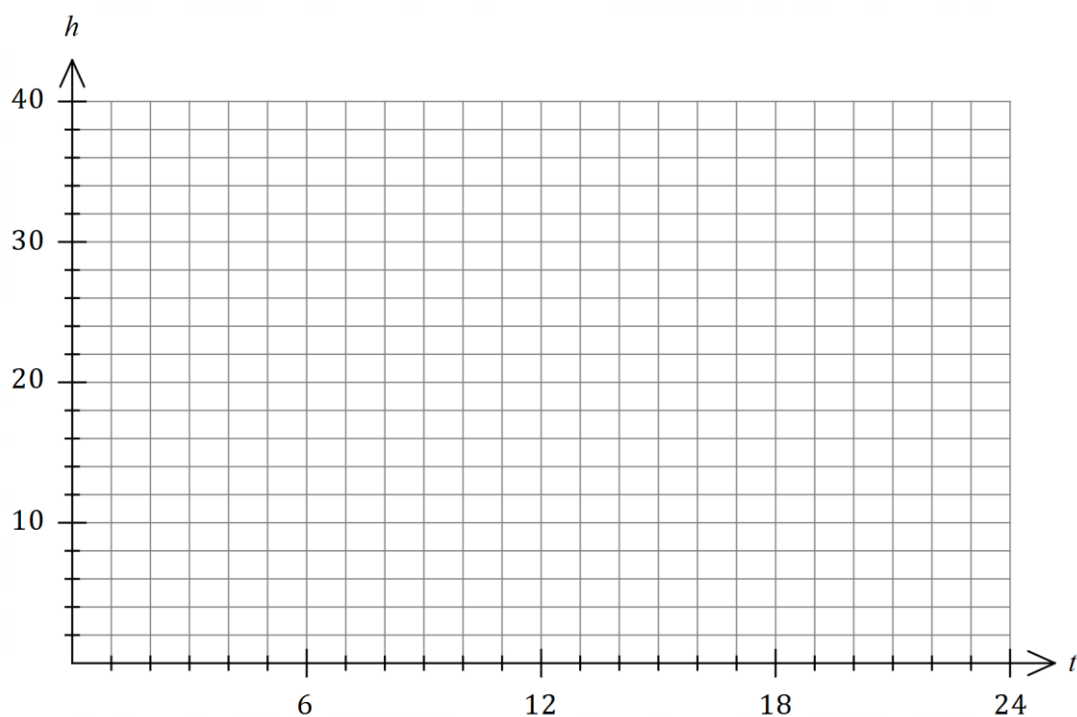


**Question 24 (8 marks)**

The height above ground level,  $h$  m, of a seat on a steadily rotating Ferris wheel  $t$  minutes after the wheel begins to move is given by  $h = 17.5 + 15.5 \cos\left(\frac{\pi t}{6} - \frac{\pi}{3}\right)$ .

- (a) Determine the initial height of the seat. (1 mark)

- (b) Graph the height of the seat against time on the axes below. (4 marks)



- (c) Determine

- (i) the maximum height above ground reached by the seat. (1 mark)

- (ii) the time taken, to the nearest second, for the seat to first reach a height of 7 m above ground level. (2 marks)

### **Additional working space**

Question number(s): .....

### **Additional working space**

Question number(s): .....